

It's not you, it's the math:
your friends are (probably) more important than you

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ABSTRACT

This study provides a comprehensive empirical and theoretical investigation of the Generalized Friendship Paradox across diverse network topologies. By evaluating global and local friendship gaps across six standard centrality metrics, we show that real-world networks have very unequal structures, with a few highly connected individuals. This high inequality strongly increases the illusion that our friends are more popular or important than we are, far more than in purely Erdős–Rényi random networks models. Furthermore, we introduce $\exp(-\Delta_i)$ as a novel localized centrality metric, mapping a node's relative standing within its immediate neighborhood and offering a complementary perspective to traditional global centrality measures.

Repository: <https://github.com/your-repository-link>

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List of Symbols

Symbol	Description
$G = (V, E)$	A simple, undirected graph with vertex set V and edge set E
$n = V $	Number of vertices (nodes) in the graph
$m = E $	Number of edges in the graph
d_i	Degree of node i , i.e., $d_i = N(i) $
$N(i)$	Open neighborhood of node i , i.e., $\{j \in V : (i, j) \in E\}$
$\bar{k} = \langle k \rangle$	Average degree of the network, $\frac{1}{n} \sum_{i=1}^n d_i$

List of Abbreviations

Abbreviation	Full Form
FP	Friendship Paradox
GFP	Generalized Friendship Paradox
ER	Erdős–Rényi (random graph model)

1 Introduction

1.1 Motivation and Context

The Friendship Paradox (FP), first articulated by Feld in 1991, states that "on average, your friends have more friends than you do" [1]. This counterintuitive phenomenon arises from a fundamental sampling bias in networks: when individuals are sampled through their connections, highly connected nodes are more likely to be encountered than nodes with fewer connections. As a result, the local view of a network can differ substantially from its global structure.

The Generalized Friendship Paradox (GFP) extends this idea beyond degree by considering other measures of node importance, including centrality measures and node attributes [2]. It shows that similar biases can arise across different dimensions of network structure, suggesting that the way a network is sampled can strongly influence how the importance of its nodes is perceived.

1.2 Practical Applications of the Friendship Paradox

While initially a curiosity in social psychology, the FP has since been recognized as a critical factor affecting diverse domains because highly connected individuals are more likely to appear when we observe the friends of randomly selected individuals.

- **Public health and epidemic control.** Monitoring the friends of randomly selected individuals can help identify highly connected people who may play an important role in disease transmission [3]. This idea can be used for early outbreak detection and for designing more efficient vaccination or monitoring strategies.
- **Information detection and diffusion.** Highly connected individuals are more likely to encounter new information early and to spread it quickly through a network [4]. Observing the friends of a random sample can therefore help detect emerging trends, products, or opinions, while targeting well-connected individuals can facilitate the rapid dissemination of important information.
- **Social media and social comparison.** The paradox also helps explain why people on social media may appear more popular or socially active than they really are [5]. Highly connected users are naturally overrepresented in other users' networks and feeds.

1.3 Research Questions

To evaluate how local perceptions differ from global network structures, we quantify the local friendship gap Δ_i , defined as the difference between the average importance of a node's immediate neighbors and its own importance. Building on this framework, this work addresses the following research questions:

- **RQ1:** How does the Generalized Friendship Paradox (GFP) behave across different node centrality measures, and how do empirical graph structures deviate from theoretical null models?
- **RQ2:** How does the local Friendship Paradox vary across real-world and random networks, and what structural properties of a network are associated with the local gap Δ_i ?
- **RQ3:** Can $\exp(-\Delta_i)$ provide a meaningful centrality measure, and how does it compare with existing centrality measures?

1.4 Research Contributions and Structure of the Work

This work provides a systematic empirical and theoretical analysis of both global and local formulations of the Friendship Paradox across real-world networks and synthetic null models. The main contributions of this paper are summarized as follows:

1. **Comprehensive Evaluation of the GFP:** We benchmark the Generalized Friendship Paradox across six distinct centrality measures (Degree, Eigenvector, Walk, Katz, Exponential, and Non-backtracking centralities) on five diverse empirical networks and their corresponding Erdős–Rényi random graphs.
2. **Characterization of the Local Friendship Gap (Δ_i):** We formalize the local friendship gap Δ_i for general importance vectors x , analyzing its empirical distribution, expected value, and proportion of paradoxical nodes ($\Delta_i \geq 0$) across networks.
3. **Analytical Null Model Derivation:** We establish theoretical normal approximations for the distribution of Δ_i in Erdős–Rényi random graphs and validate them empirically.
4. **Novel Centrality Metric $\exp(-\Delta_i)$:** We introduce $\exp(-\Delta_i)$ as a localized centrality measure reflecting relative node standing within immediate neighborhoods and analyze its correlation with traditional global centralities.

The remainder of this report is organized as follows: Section 2 presents on the friendship paradox. Section 3 provides the mathematical foundations and graph notation. Section 4 details the datasets and experimental methodology. Section 5 presents the global framework of the Friendship Paradox. Section 6 analyzes the local gap Δ_i . Section 7 evaluates the proposed centrality metric $\exp(-\Delta_i)$, and Section 8 concludes the work. Additional figures and empirical distributions are compiled in the Appendix.

2 Related Work

The Friendship Paradox, introduced by Feld [1], established that an individual’s friends have, on average, more friends than the individual themselves due to sampling bias toward highly connected nodes. Over the past decade, this foundational concept has been significantly expanded to investigate whether similar biases persist for non-degree attributes, leading to the mathematical formulation of the Generalized Friendship Paradox [6, 2].

2.1 Empirical Foundations

Eom and Jo [6] were among the first to formalize the GFP for arbitrary node characteristics across real-world scientific collaboration networks. They demonstrated that the GFP holds at both individual and aggregate levels for non-degree characteristics such as citation counts, publication volume, and co-author numbers. Their work established that the mathematical origin of the GFP stems directly from positive correlations between node degree and target characteristics, and suggested leveraging the GFP for efficient sampling strategies to identify highly influential nodes in large networks.

2.2 Analytical Models

Extending beyond mean behavior, Cantwell et al. [7] developed a comprehensive mathematical framework for both the classical and generalized friendship paradoxes across random network models and empirical datasets. Using generating function methods, they derived the full probability distributions of neighbor degree differences, accounting for variance around expected values. Their empirical comparisons across a vast collection of real-world datasets confirmed strong agreement with theoretical null model predictions while emphasizing the role of network structure in driving local variance.

2.3 Linear Algebra and Centrality Extensions

Most recently, Higham et al. [2] derived new theoretical bounds for the GFP using a linear algebra perspective. Comparing global (degree-weighted) and local neighborhood averages, they proved that a global GFP holds for a broad family of walk-based centralities, such as Katz centrality and total subgraph communicability, as well as non-backtracking eigenvector centrality. However, they identified counterexamples for centralities based strictly on even-length walks. For local averaging, they characterized exact conditions where the local paradox holds and demonstrated that for specific inverse attributes (such as loneliness, defined as $1/d_i$), the paradox holds in reverse.

Building upon these foundational insights, our work systematically evaluates global and local GFP formulations across real-world networks and Erdős–Rényi synthetic graphs, while introducing $\exp(-\Delta_i)$ to capture localized neighborhood standing.

3 Mathematical Background

3.1 Graph Notation

Let us consider a graph with n nodes and m edges. We denote the adjacency matrix of this graph as A . Let $\mathbf{1} \in \mathbb{R}^n$ be the vector of all ones. The degree vector of the graph is defined as $d = A\mathbf{1}$, and we define $D = \text{diag}(d)$ as the diagonal matrix containing the node degrees. Furthermore, let $x \in \mathbb{R}^n$ represent a generic vector of node importances or attributes.

3.2 Centrality Measures

To evaluate the structural importance of nodes within the network, several centrality measures can be defined based on the adjacency matrix A . Each metric captures a different perspective on what makes a node "central" or influential.

3.2.1 Degree Centrality

The degree vector is defined as:

$$d = A\mathbf{1} \tag{1}$$

Key Insight. This is the simplest measure of centrality. It merely counts the number of direct neighbors a node possesses. In a social network context, it represents immediate popularity or the direct reach a person has without relying on intermediaries.

3.2.2 Eigenvector Centrality

Eigenvector centrality assigns relative scores to all nodes in the network based on the principle that connections to high-scoring nodes contribute more to the score of the node in question. Mathematically, it corresponds to the leading eigenvector x associated with the largest eigenvalue $\lambda_{\max}$ of the adjacency matrix A :

$$Ax = \lambda_{\max}x \implies x_i = \frac{1}{\lambda_{\max}} \sum_{j=1}^n A_{ij}x_j \tag{2}$$

where x is normalized and guaranteed to have non-negative entries by the Perron–Frobenius theorem.

Key Insight. Unlike degree centrality, which treats all connections equally, eigenvector centrality operates on the principle that connections to highly influential nodes contribute more to a node's own influence than connections to less influential nodes. It is a recursive measure where a node is important if its friends are also important.

3.2.3 Walk Centrality

Given by the equation:

$$x = A^\ell \mathbf{1} \tag{3}$$

Key Insight. Walk centrality counts the exact number of walks of length l that originate from a given node. It measures the ability of a node to spread or gather information exactly l steps away, capturing a bounded sphere of influence.

3.2.4 Katz Centrality

Computed by solving the linear system:

$$(I - \alpha A)x = \mathbf{1} \implies x = (I - \alpha A)^{-1}\mathbf{1} = \sum_{k=0}^{\infty} \alpha^k A^k \mathbf{1} \quad (4)$$

where $0 < \alpha < 1/\lambda_{\max}(A)$ is an attenuation parameter.

Key Insight. Katz centrality extends the concept of walk centrality by counting walks of all possible lengths, applying an attenuation factor α to penalize longer walks. It grants a baseline centrality (the vector $\mathbf{1}$) to all nodes, meaning even a node with no incoming links maintains a minimal level of importance.

3.2.5 Total Subgraph Communicability (Exponential Centrality)

Defined using the matrix exponential as:

$$x = \exp(\alpha A)\mathbf{1} = \sum_{k=0}^{\infty} \frac{\alpha^k A^k}{k!} \mathbf{1} \quad (5)$$

Key Insight. Similar to Katz centrality, this measure considers walks of all lengths. However, it penalizes longer walks much more aggressively by dividing by the factorial of the walk length $k!$. It heavily emphasizes a node's participation in short, dense, local substructures.

3.2.6 Nonbacktracking Eigenvector Centrality

Corresponds to the vector formed by taking the first n entries (normalized to have an L_2 -norm of 1) of the leading eigenvector v of the block matrix B :

$$\begin{bmatrix} A & I - D \\ I & 0 \end{bmatrix} v = \lambda_{\max} v, \quad \text{with } x = \frac{v_{1:n}}{\|v_{1:n}\|_2} \quad (6)$$

Key Insight. Standard walk-based centralities suffer from "echo chambers" or localization, where a walk simply bounces back and forth between two highly connected nodes (hubs), artificially inflating their importance. Nonbacktracking centrality prevents a walk from immediately returning to the node it just came from, providing a more robust measure of influence, especially in sparse networks or networks with massive hubs.

3.3 Random Graphs: Erdős–Rényi

To study whether the structural patterns observed in real networks differ from purely random configurations, we utilize the Erdős–Rényi random graph model $G(n, p)$. In this work, we choose p such that the synthetic random graph has, in expectation, the exact same number of edges as the empirical target graph. The derivation of p proceeds as follows:

1. **Total Potential Edges:** In a simple undirected graph with n nodes, each pair of nodes can potentially form an edge. The total number of possible pairs is given by the binomial coefficient:

$$\binom{n}{2} = \frac{n(n-1)}{2} \quad (7)$$

In $G(n, p)$, each potential edge is present independently with probability p .

2. **Expected Number of Edges:** Let M denote the total number of edges in the random graph. We can express M as a sum of indicator random variables:

$$M = \sum_{1 \leq i < j \leq n} X_{ij} \quad (8)$$

where X_{ij} is defined as:

$$X_{ij} = \begin{cases} 1 & \text{if edge } (i, j) \text{ exists} \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

Since $X_{ij} \sim \text{Bernoulli}(p)$, its expected value is $\mathbb{E}[X_{ij}] = p$. By the linearity of expectation, the expected total number of edges is:

$$\mathbb{E}[M] = \sum_{1 \leq i < j \leq n} \mathbb{E}[X_{ij}] = \binom{n}{2} p = \frac{n(n-1)}{2} p \quad (10)$$

3. **Matching the Real Graph's Density:** Suppose our real-world graph possesses m edges. We require the expected number of edges in our random model to match this value, meaning $\mathbb{E}[M] = m$. Thus:

$$\frac{n(n-1)}{2} p = m \quad (11)$$

Isolating the probability p yields:

$$p = \frac{2m}{n(n-1)} \quad (12)$$

4 Datasets and Experimental Methodology

4.1 Real-World Network Datasets

To empirically evaluate the behavior of the Friendship Paradox and our proposed local metrics across diverse topological structures, we select a representative collection of real-world networks from various domains. Table 1 summarizes the main topological properties of these empirical networks, including the number of nodes (n), number of edges (m), average degree ($\bar{k}$), and density (p).

Table 1: Summary statistics of empirical real-world networks.

Dataset	Nodes (n)	Edges (m)	Avg. Degree ($\bar{k}$)	Density (p)
Karate Club [8]	34	78	4.59	0.1390
Dolphins [9]	15	20	2.67	0.1905
Les Misérables [10]	77	254	6.60	0.0868
Cora [11]	2708	5278	3.90	0.0014
CiteSeer [12]	3327	4552	2.74	0.0008

Description of Topological Characteristics. As summarized in Table 1, these metrics allow us to compare the scale and global connectivity of each network:

- **Number of Nodes (n):** Represents the total size of the network by counting individual elements (e.g., individuals in a social network like *Karate Club*, or academic papers in *Cora* and *CiteSeer*).
- **Number of Edges (m):** Quantifies the total number of connections or interactions present between pairs of nodes (e.g., friendship ties, co-appearances, or citation links).
- **Average Degree ($\bar{k}$):** Measures the average number of connections per node, computed as $\bar{k} = \frac{2m}{n}$. As shown in Table 1, *Les Misérables* exhibits the highest average degree ($\bar{k} = 6.60$), reflecting tightly grouped interactions around central characters.
- **Density (p):** Reflects the proportion of actual edges relative to the maximum possible number of edges, calculated via $p = \frac{2m}{n(n-1)}$. Table 1 illustrates the clear contrast between small, highly dense social networks (such as *Dolphins* with $p = 0.1905$) and large, extremely sparse citation networks (such as *CiteSeer* with $p = 0.0008$).

4.2 Synthetic Network Generation

For each empirical dataset, we generate corresponding null models using the Erdős–Rényi random graph model $G(n, p)$. As derived in Section 2, the edge creation probability p is calibrated directly to preserve the expected density and total number of edges of the empirical network:

$$p = \frac{2m}{n(n-1)} \quad (13)$$

Comparing real-world topologies with these matched Erdős–Rényi graphs allows us to isolate the specific effects of non-random network structures (such as heavy-tailed degree distributions and community structure) on the Friendship Paradox.

5 The Global Framework

5.1 Global Friendship Paradox

Using the generic vector of importances $\mathbf{x} \in \mathbb{R}^n$, the Global Friendship Paradox can be mathematically expressed as an inequality comparing the average centrality of a randomly chosen node to that of a randomly chosen neighbor across the network.

5.1.1 Mathematical Computation of Expected Values

1. **Node-Level Expected Centrality ($\mathbb{E}[x]$):** The expected centrality of a node chosen uniformly at random from the graph is given by the unweighted global mean:

$$\mathbb{E}[x] = \frac{\mathbf{1}^\top \mathbf{x}}{n} = \frac{1}{n} \sum_{i=1}^n x_i \quad (14)$$

2. **Neighbor-Level Expected Centrality ($\mathbb{E}[x_{\text{neigh}}]$):** When picking a random edge and following it to a node, the probability of selecting node i is proportional to its degree d_i . Thus, the expected centrality of a randomly chosen neighbor is represented by the degree-weighted average of the node importances:

$$\mathbb{E}[x_{\text{neigh}}] = \frac{\mathbf{d}^\top \mathbf{x}}{\mathbf{1}^\top \mathbf{d}} = \frac{1}{2m} \sum_{i=1}^n d_i x_i \quad (15)$$

3. **Global Inequality and Gap (Δ_{GFP}):** The Global Friendship Paradox states that the degree-weighted average is greater than or equal to the unweighted global average:

$$\frac{\mathbf{d}^\top \mathbf{x}}{\mathbf{1}^\top \mathbf{d}} \geq \frac{\mathbf{1}^\top \mathbf{x}}{n} \quad (16)$$

This reflects the fundamental sampling bias inherent in network connectivity: high-degree nodes are overrepresented when sampling through edges. The magnitude of this effect is quantified by the global gap:

$$\Delta_{\text{GFP}} = \mathbb{E}[x_{\text{neigh}}] - \mathbb{E}[x] \quad (17)$$

5.2 Global Paradox Comparison in Real-world and Random Graphs

We examine the magnitude of Δ_{GFP} across empirical networks and their Erdős–Rényi baseline counterparts $G(n, p)$, where $p = \frac{2m}{n(n-1)}$ as derived in Section 3.3.

Table 2 presents the empirical results for the Global Friendship Paradox across all six centrality measures for the first two networks (*Karate Club* and *Dolphins*). The results for the remaining datasets are provided in Appendix A.

As shown in Table 2, the global GFP inequality $\mathbb{E}[x_{\text{neigh}}] \geq \mathbb{E}[x]$ holds across virtually all real-world networks and centralities. However, the magnitude of the global gap Δ_{GFP} is consistently higher in real networks than in Erdős–Rényi graphs. For instance, in *Karate Club* under Walk centrality, $\Delta_{\text{GFP}} = 140.5354$ in the real graph compared to only 3.4336 in the random model. This discrepancy is directly driven by the higher degree variance and structural heterogeneity observed in real-world topologies.

Table 2: Global Friendship Paradox evaluation across networks and centralities.

Graph	Centrality	Real $\mathbb{E}[x]$	Real $\mathbb{E}[x_{\text{neigh}}]$	Real Δ_{GFP}	Random $\mathbb{E}[x]$	Random $\mathbb{E}[x_{\text{neigh}}]$	Random Δ_{GFP}
Karate Club	Degree	0.1390	0.2354	0.0964	0.1444	0.1624	0.0180
Karate Club	Eigenvector	0.1464	0.2146	0.0682	0.1543	0.1787	0.0244
Karate Club	Walk	320.8235	461.3590	140.5354	25.5294	28.9630	3.4336
Karate Club	Katz	4.6542	6.6852	2.0310	5.6663	6.3580	0.6917
Karate Club	Exponential	1.1531	1.2556	0.1025	1.0489	1.0551	0.0061
Karate Club	Non-backtracking	0.1378	0.2121	0.0742	-0.1596	-0.1800	-0.0204
Dolphins	Degree	0.1905	0.2393	0.0488	0.1810	0.2331	0.0521
Dolphins	Eigenvector	0.2335	0.2851	0.0516	0.2280	0.2887	0.0607
Dolphins	Walk	8.9333	10.5000	1.5667	8.2667	10.2105	1.9439
Dolphins	Katz	5.6537	6.6851	1.0314	5.4543	6.6341	1.1798
Dolphins	Exponential	1.0271	1.0340	0.0069	1.0258	1.0331	0.0074
Dolphins	Non-backtracking	0.2421	0.2783	0.0363	0.2409	0.2860	0.0451

6 The Local Friendship Gap Δ_i

6.1 Definition

For a given node i , the local friendship gap Δ_i measures the relative perception bias experienced locally by node i :

$$\Delta_i = (D^{-1}Ax)_i - x_i \quad (18)$$

6.2 Interpretation of Δ_i

When $\Delta_i > 0$, node i suffers from the local paradox: on average, its immediate neighbors possess a higher centrality score than node i itself. Conversely, $\Delta_i < 0$ indicates a dominant node whose neighbors are less central on average than itself.

6.3 Empirical distribution of Δ_i

The continuous density distributions of Δ_i across networks highlight distinct behavior between empirical topologies and random networks. Figure 1 illustrates the density of Δ_i for the *Dolphins* network across six centralities.

As seen in Figure 1, because *Dolphins* is a small, relatively homogeneous network ($\text{Var}(d) = 1.82$), the distributions for the real network (red) and the Erdős-Rényi graph (blue) closely mirror each other. In both cases, the peak of the distribution lies to the right of $\Delta_i = 0$, confirming that a majority of nodes experience $\Delta_i \geq 0$ (80.0% in real vs 64.29% in random for degree centrality).

Table 3 summarizes the expected local gap $\mathbb{E}[\Delta_i]$ and the percentage of paradoxical nodes ($\%\Delta_i \geq 0$) for representative networks.

Table 3: Summary of local gap metrics $\mathbb{E}[\Delta_i]$ and $\%\Delta_i \geq 0$ across real and random networks.

Graph	Centrality	Real $\%\Delta_i \geq 0$	Random $\%\Delta_i \geq 0$	Real $\mathbb{E}[\Delta_i]$	Random $\mathbb{E}[\Delta_i]$
Karate Club	Degree	85.29%	64.71%	0.1522	0.0164
Karate Club	Exponential	85.29%	64.71%	0.1581	0.0055
Les Miserables	Degree	87.01%	63.64%	0.0928	0.0095
Les Miserables	Non-backtracking	85.71%	28.57%	0.0868	-0.0098
Cora	Degree	84.79%	67.40%	0.0029	0.0003
Cora	Eigenvector	96.82%	80.16%	0.0126	0.0035
CiteSeer	Degree	81.52%	70.63%	0.0008	0.0002
CiteSeer	Eigenvector	92.16%	71.76%	0.0053	0.0028

In larger real-world networks such as *Cora* and *CiteSeer*, the proportion of paradoxical nodes under eigenvector centrality reaches extreme levels (96.82% for *Cora* and 92.16% for *CiteSeer*, as detailed in Table 3).

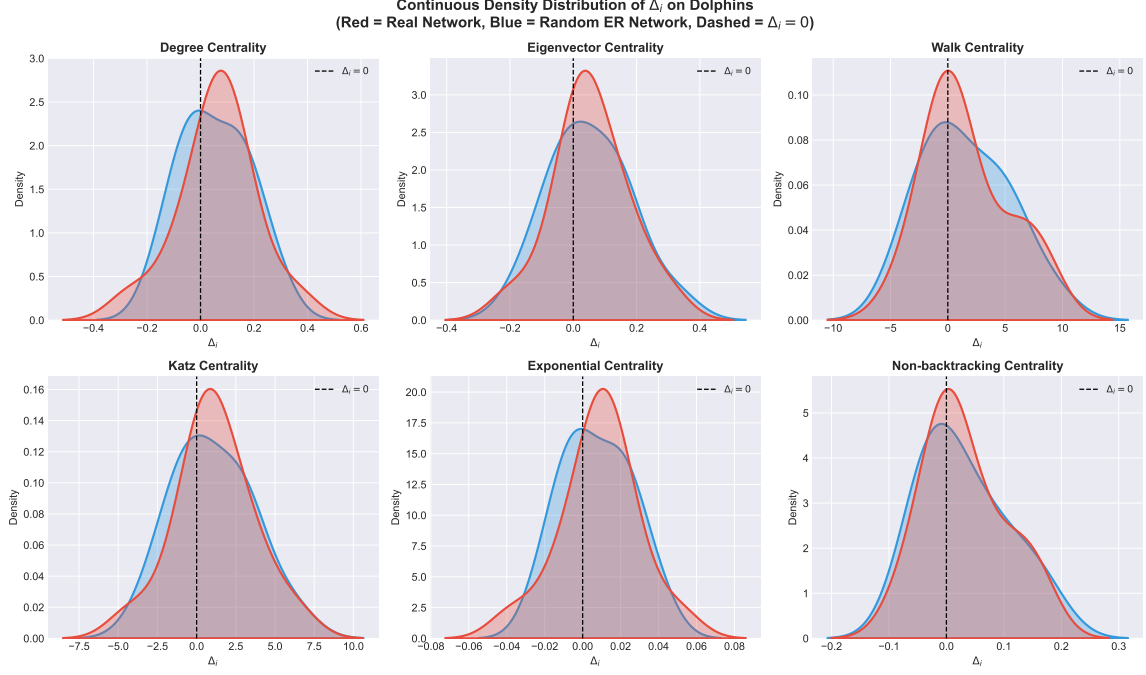


Figure 1: Continuous density distribution of local gap Δ_i on the *Dolphins* network across six centrality measures (Red: Real Network, Blue: Random ER Network, Dashed line: $\Delta_i = 0$).

6.4 Δ_i versus degree and baseline attribute vector $\vec{x}$

Across all tested datasets, Δ_i exhibits a strong inverse relationship with node degree d_i and baseline importance x_i . Highly connected hub nodes routinely display negative values ($\Delta_i < 0$) because their neighbors consist primarily of peripheral nodes with lower average degree. Conversely, low-degree nodes are almost exclusively connected to higher-degree neighbors, leading to positive local gaps ($\Delta_i > 0$).

7 A New Centrality: $\exp(-\Delta_i)$

We propose a local centrality metric defined as:

$$C_{\text{local}}(i) = \exp(-\Delta_i) \quad (19)$$

The exponential transformation maps $\Delta_i \in (-\infty, +\infty)$ onto $(0, +\infty)$. Nodes that significantly outperform their local neighborhoods ($\Delta_i \ll 0$) receive high scores ($C_{\text{local}} \gg 1$), while locally disadvantaged nodes ($\Delta_i > 0$) receive scores bounded in $(0, 1)$.

To evaluate whether $\exp(-\Delta_i)$ captures structural prominence comparable to standard global centralities x , we analyze the Pearson correlation coefficient $\text{Corr}(\exp(-\Delta_i), x)$ reported in Table 4.

Table 4: Pearson correlation $\text{Corr}(\exp(-\Delta_i), x)$ between proposed local centrality and original centrality x .

Graph	Degree	Eigenvector	Walk	Katz	Exponential	Non-backtracking
Karate Club	0.9363	0.8069	0.3132	0.5572	0.9405	0.8443
Dolphins	0.9034	0.8326	0.4250	0.5491	0.8902	0.8087
Les Misérables	0.8101	0.6030	–	0.4976	0.8467	0.7332
Cora	0.3546	-0.2627	0.0000	0.6607	0.5762	-0.2628
CiteSeer	0.2812	-0.1309	0.0000	0.4820	0.3803	-0.1396

As demonstrated in Table 4, $\exp(-\Delta_i)$ exhibits strong positive alignment with degree and exponential centralities in social networks (e.g., $r = 0.9405$ for Exponential Centrality in *Karate Club*). In sparse citation networks like *Cora* and *CiteSeer*, local neighborhood dynamics diverge from global eigenvector centrality ($r = -0.2627$), highlighting that $\exp(-\Delta_i)$ provides a distinct, locally grounded perspective on node importance.

8 Conclusion and Future Work

In this work, we presented a comprehensive examination of the Generalized Friendship Paradox across empirical and synthetic networks. We demonstrated that structural heterogeneity and degree variance in real-world graphs amplify both global and local friendship gaps compared to Erdős–Rényi random graphs. Furthermore, we introduced $\exp(-\Delta_i)$ as an effective localized metric for identifying relative structural dominance within immediate network neighborhoods. Future research will explore the behavior of local friendship gaps in dynamic and multi-layer networks.

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A Extended Experimental Results of the Global Friendship Paradox

This appendix presents the complete experimental results for the Global Friendship Paradox evaluation across the remaining real-world network datasets: *Les Miserables*, *Cora*, and *CiteSeer*.

Table 5 details the empirical expected centrality values $\mathbb{E}[x]$, neighbor expected centralities $\mathbb{E}[x_{\text{neigh}}]$, and the resulting paradox gaps Δ_{GFP} compared against equivalent Erdős–Rényi random models $G(n, p)$.

Table 5: Global Friendship Paradox evaluation for additional real-world networks and centralities.

Graph	Centrality	Real $\mathbb{E}[x]$	Real $\mathbb{E}[x_{\text{neigh}}]$	Real Δ_{GFP}	Random $\mathbb{E}[x]$	Random $\mathbb{E}[x_{\text{neigh}}]$	Random Δ_{GFP}
Les Miserables	Degree	0.0868	0.1586	0.0718	0.0844	0.0934	0.0090
Les Miserables	Eigenvector	0.0780	0.1420	0.0640	0.1053	0.1181	0.0129
Les Miserables	Walk	1220.8831	2552.5591	1331.6759	45.5325	50.1984	4.6659
Les Miserables	Katz	2.5664	4.4042	1.8378	5.8980	6.4659	0.5678
Les Miserables	Exponential	1.2885	1.6007	0.3122	1.0665	1.0735	0.0071
Les Miserables	Non-backtracking	0.0584	0.1294	0.0710	-0.1076	-0.1187	-0.0111
Cora	Degree	0.0014	0.0040	0.0026	0.0014	0.0018	0.0004
Cora	Eigenvector	0.0048	0.0177	0.0129	0.0153	0.0202	0.0049
Cora	Walk	42.5251	83.5784	41.0533	18.6839	23.4869	4.8030
Cora	Katz	1.7183	3.1200	1.4016	4.8549	5.9866	1.1317
Cora	Exponential	1.0412	1.1135	0.0723	1.0395	1.0497	0.0102
Cora	Non-backtracking	0.0075	0.0194	0.0118	0.0166	0.0209	0.0042
CiteSeer	Degree	0.0008	0.0021	0.0013	0.0008	0.0011	0.0003
CiteSeer	Eigenvector	0.0034	0.0170	0.0136	0.0093	0.0140	0.0047
CiteSeer	Walk	18.9179	54.3926	35.4746	9.6327	13.0626	3.4299
CiteSeer	Katz	1.4204	2.4843	1.0639	3.8667	5.0678	1.2012
CiteSeer	Exponential	1.0283	1.0720	0.0436	1.0271	1.0368	0.0097
CiteSeer	Non-backtracking	0.0036	0.0164	0.0128	-0.0140	-0.0189	-0.0049

B Annexe A: Comparative Density Plots of Δ_i

This section presents the continuous density distributions of Δ_i across six centrality measures for the *Karate Club*, *Les Miserables*, and *Cora* networks.

Figure 2 shows the density distribution for the *Karate Club* network. In this highly heterogeneous small network ($\text{Var}(d) = 14.60$), the real graph distribution (red) exhibits a significantly wider spread and higher mean ($\mathbb{E}[\Delta_i] = 0.1522$) compared to the random model ($\mathbb{E}[\Delta_i] = 0.0164$).

Figure 3 displays the density distribution for *Les Miserables*.

Figure 4 illustrates the distribution for the large, sparse *Cora* citation network.

C Annexe B: Theoretical Normal Fits for Erdős–Rényi Null Models

In Erdős–Rényi random graphs, the distribution of Δ_i under degree centrality can be modeled theoretically using a Gaussian normal fit $N(\mu, \sigma^2)$. Figure 5 and Figure 6 compare the theoretical a priori normal fit against the empirical random graph distributions for *Karate Club* and *CiteSeer*.

As shown in Figure 5 and Figure 6, the empirical distributions of Δ_i in synthetic null models match the theoretical normal approximations closely, validating our analytical framework.

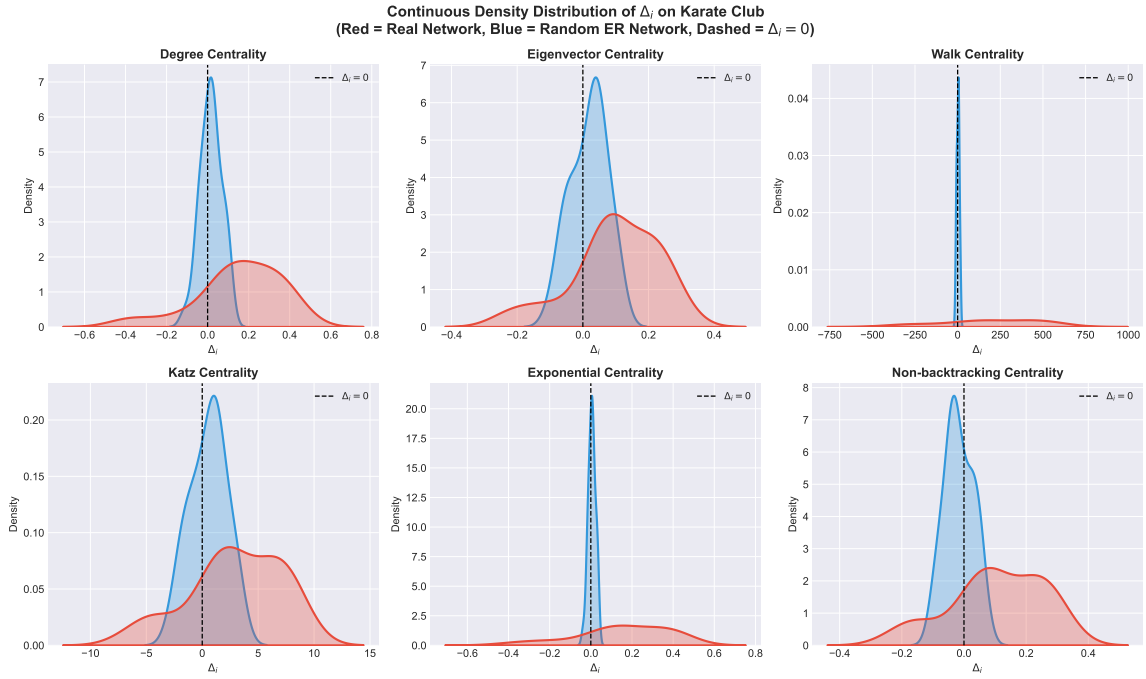


Figure 2: Continuous density distribution of Δ_i on *Karate Club* (Red: Real Network, Blue: Random ER Network, Dashed line: $\Delta_i = 0$).

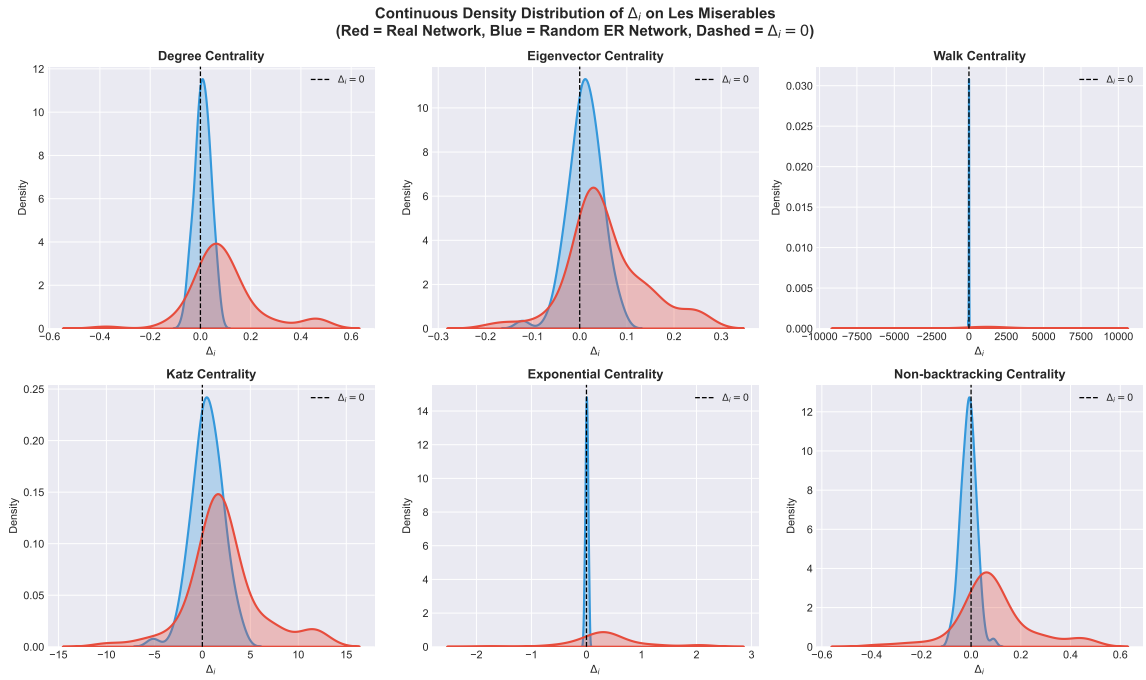


Figure 3: Continuous density distribution of Δ_i on *Les Miserables* (Red: Real Network, Blue: Random ER Network, Dashed line: $\Delta_i = 0$).

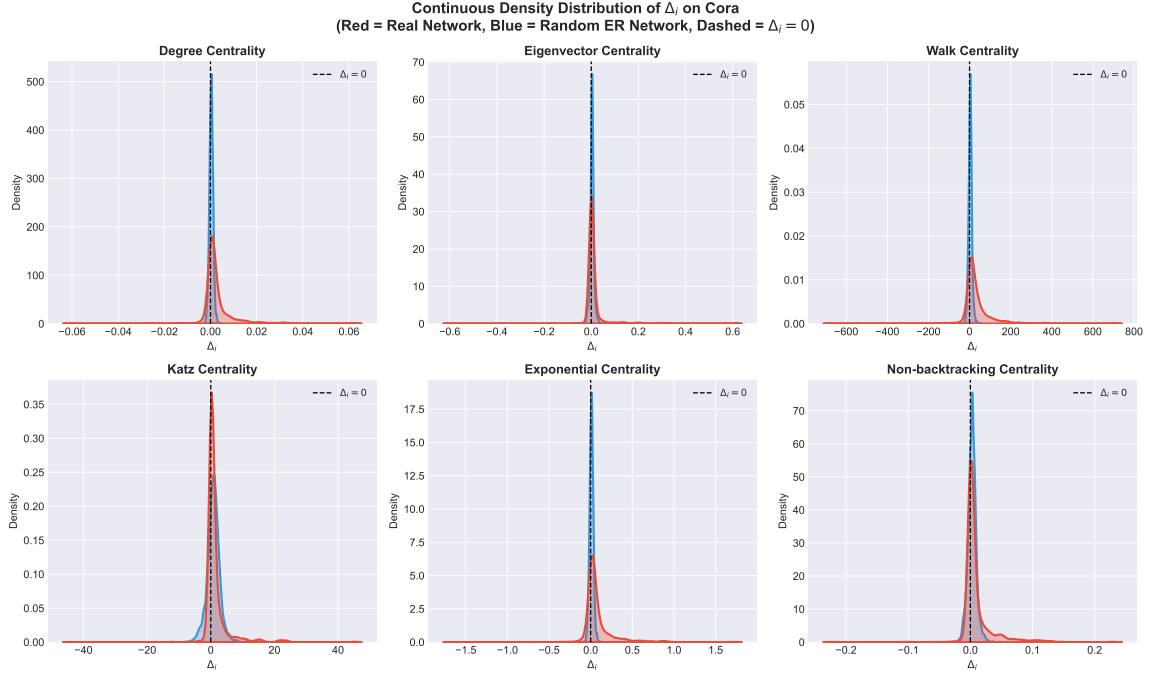


Figure 4: Continuous density distribution of Δ_i on *Cora* (Red: Real Network, Blue: Random ER Network, Dashed line: $\Delta_i = 0$).

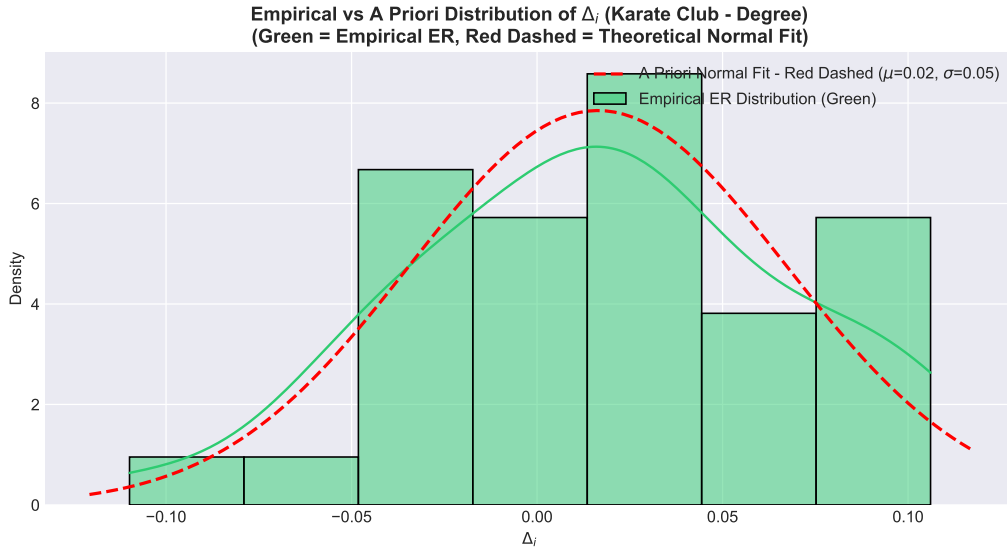


Figure 5: Empirical ER distribution vs. theoretical A Priori Normal Fit ($\mu = 0.02, \sigma = 0.05$) of Δ_i for *Karate Club* under Degree Centrality.

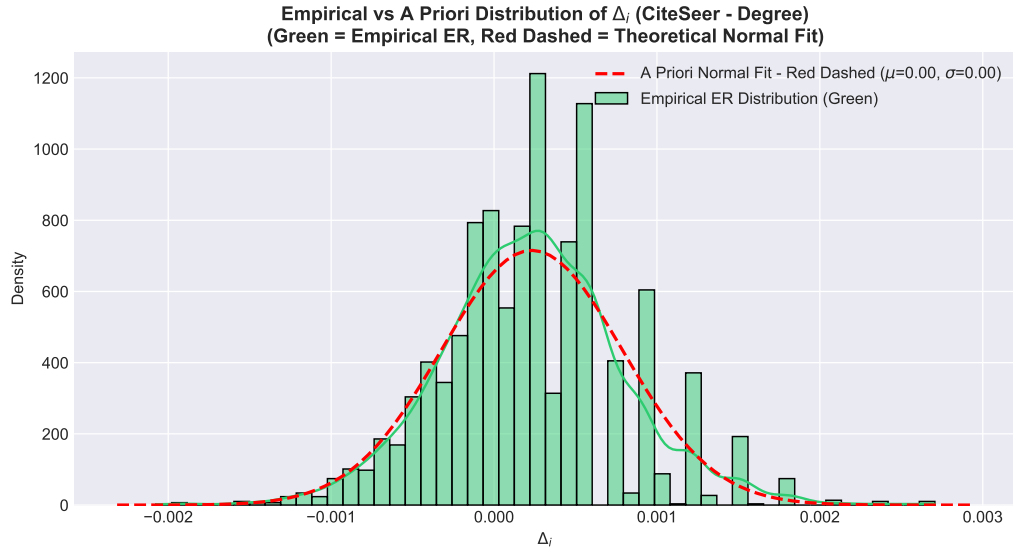


Figure 6: Empirical ER distribution vs. theoretical A Priori Normal Fit ($\mu = 0.00, \sigma = 0.00$) of Δ_i for *CiteSeer* under Degree Centrality.